



LOAN APPROVAL

EXPLORATRY
DATA
ANALYSIS



Data analytics

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About the loan

In today's financial world, loan approval is one of the most important processes for both banks and applicants. Every bank receives thousands of loan applications, but not all are approved. The approval of a loan depends on several factors such as applicant's income, employment status, credit history, and other demographic details.

This project focuses on analyzing the factors that influence loan approval decisions using a dataset containing features like Applicant Income, Coapplicant Income, Loan Amount, Loan Term, Credit History, Education, Property Area, and more. By studying these variables, we can understand the pattern behind approved and rejected loans.

The main goal of this study is to build a data-driven model that can predict whether a loan application will be approved or not, based on the applicant's financial and personal information. This helps financial institutions make faster and more reliable decisions while ensuring fairness and efficiency in the loan approval process.

Description

- Loading and Preparing **Loan approval** Dataset for Analysis in Google Colab

```
[ ] from google.colab import drive  
    drive.mount('/content/drive')
```

Mounted at /content/drive

- in this analysis , we import **Pandas** for data manipulation ,**Numpy** for numerical operations ,**Matplotlib** for creating visualization ,and **Seaborn** and **plotly** for enhanced statistical graphic.

```
import numpy as np  
import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt  
import plotly.express as px
```

- We import the Google drive module to mount Google `Drive`, enabling access to files and dataset in the for our analysis.

```
df=pd.read_csv("/content/drive/MyDrive/Car vehicle/loan_sanction_test.csv")
```

- We use pandas to Read the CSV file containing big vehicle insurance from Google Drive, loading it into a DATAFRAME Name 'df' for analysis.

Data overview

df												
	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area
0	LP001015	Male	Yes	0	Graduate	No	5720	0	110.0	360.0	1.0	Urban
1	LP001022	Male	Yes	1	Graduate	No	3076	1500	126.0	360.0	1.0	Urban
2	LP001031	Male	Yes	2	Graduate	No	5000	1800	208.0	360.0	1.0	Urban
3	LP001035	Male	Yes	2	Graduate	No	2340	2546	100.0	360.0	NaN	Urban
4	LP001051	Male	No	0	Not Graduate	No	3276	0	78.0	360.0	1.0	Urban
...	...	...	...	...	...	...	...	...	...	...	...	...
362	LP002971	Male	Yes	3+	Not Graduate	Yes	4009	1777	113.0	360.0	1.0	Urban
363	LP002975	Male	Yes	0	Graduate	No	4158	709	115.0	360.0	1.0	Urban
364	LP002980	Male	No	0	Graduate	No	3250	1993	126.0	360.0	NaN	Semiurban
365	LP002986	Male	Yes	0	Graduate	No	5000	2393	158.0	360.0	1.0	Rural
366	LP002989	Male	No	0	Graduate	Yes	9200	0	98.0	180.0	1.0	Rural

367 rows x 12 columns

- The dataset comprises of 367 rows and 12 columns.

```
f"Rows:{df.shape[0]},Columns:{df.shape[1]}"  
  
"Rows:367,Columns:12"
```

- The data set drop duplicate columns

```
df.drop_duplicates()
```

Description

The given loan dataset contains 367 records and 12 columns, representing applicant and loan-related details that can be used to predict loan approval decisions. There are no missing values, and the dataset includes both categorical (7) and numerical (5) variables. The categorical features describe demographic and employment details, while the numerical features represent income, loan, and credit information.

- **Loan_ID:** A unique identifier assigned to each loan application.
- **Gender:** Specifies the applicant's gender (Male/Female).
- **Married:** Indicates marital status (Yes/No)
- **Dependents:** Number of dependents supported by the applicant (0, 1, 2, 3+)
- **Education:** Applicant's education level (Graduate/Not Graduate).
- **Self_Employed:** Employment type, showing whether the applicant is self-employed or not.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID               367 non-null   object
1   Gender                367 non-null   object
2   Married               367 non-null   object
3   Dependents            367 non-null   object
4   Education             367 non-null   object
5   Self_Employed         367 non-null   object
6   ApplicantIncome       367 non-null   int64
7   CoapplicantIncome     367 non-null   int64
8   LoanAmount            367 non-null   int64
9   Loan_Amount_Term      367 non-null   int64
10  Credit_History         367 non-null   int64
11  Property_Area         367 non-null   object
dtypes: int64(5), object(7)
memory usage: 34.5+ KB
```

Description

- **ApplicantIncome:** The main applicant’s monthly income (numeric).
- **CoapplicantIncome:** Income of the co-applicant, if any.
- **LoanAmount:** The total amount of loan requested.
- **Loan_Amount_Term:** The duration (in months) for which the loan is taken.
- **Credit_History:** Indicates credit reliability (1 for good record, 0 for poor/no record).
-
- **Property_Area:** Type of area where the property is located (Urban, Semi-Urban, Rural).

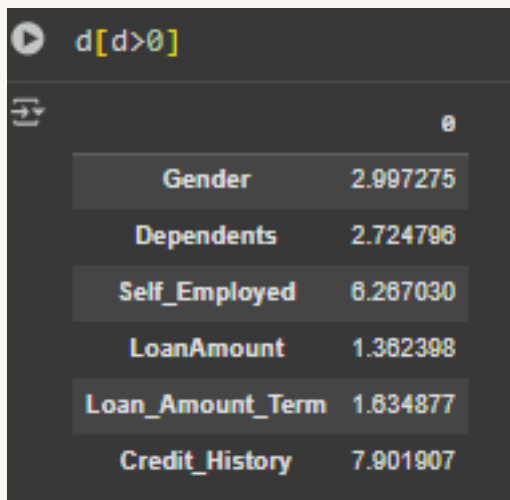
Describe

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	367.000000	367.000000	362.000000	361.000000	338.000000
mean	4805.599455	1569.577657	136.132597	342.537396	0.825444
std	4910.685399	2334.232099	61.366652	65.156643	0.380150
min	0.000000	0.000000	28.000000	6.000000	0.000000
25%	2864.000000	0.000000	100.250000	360.000000	1.000000
50%	3786.000000	1025.000000	125.000000	360.000000	1.000000
75%	5060.000000	2430.500000	158.000000	360.000000	1.000000
max	72529.000000	24000.000000	550.000000	480.000000	1.000000

Describe

Feature	Description & Insights
ApplicantIncome	The average applicant income is around ₹4806 , with a wide range (min ₹0 to max ₹72,529). The large standard deviation (4910.68) shows high income variability among applicants — some applicants earn significantly more than others.
CoapplicantIncome	Mean coapplicant income is ₹1569.57 , ranging from ₹0 to ₹24,000. Nearly 25% of applicants have no coapplicant income, suggesting that many applied individually.
LoanAmount	Average loan amount requested is ₹136.13 (in thousands) , with a minimum of ₹28,000 and a maximum of ₹550,000. The distribution is moderately spread (std = 61.37), indicating varied loan demands.
Loan_Amount_Term	The average loan term is 342.5 months (about 28.5 years) , but most applicants (50%, 75%) have terms of 360 months (30 years) . Minimum term is only 6 months , which is likely for short-term or personal loans.
Credit_History	The mean credit history score is 0.825 , suggesting that around 82.5% of applicants have a good credit history (value = 1). Minimum value of 0 indicates some applicants lack or have poor credit records.

Data Cleaning & Pre-Processing



The screenshot shows a Jupyter Notebook cell with the command `d[d>0]` and a summary table of statistics for various features. The table has two columns: the feature name and its corresponding mean value.

	0
Gender	2.997275
Dependents	2.724798
Self_Employed	6.267030
LoanAmount	1.362398
Loan_Amount_Term	1.634877
Credit_History	7.901907

We fill the missing data in the Gender ,Dependents,Self_Employed, LoanAmount,Loan_Amount_Term and Credit_History

```
df["Gender"].mode()
```

```
df['Gender'].fillna(df["Gender"].mode()[0],inplace=True)
```

```
df["Dependents"]=df["Dependents"].str.replace("+"," ")
```

```
df["Dependents"].fillna('0',inplace=True)
```

```
df["Self_Employed"].mode()
```

```
df["Self_Employed"].fillna("No",inplace=True)
```

```
df["LoanAmount"].mean()
```

```
np.float64(136.13259668508286)
```

```
df["LoanAmount"].fillna(df["LoanAmount"].mean(),inplace=True)
```

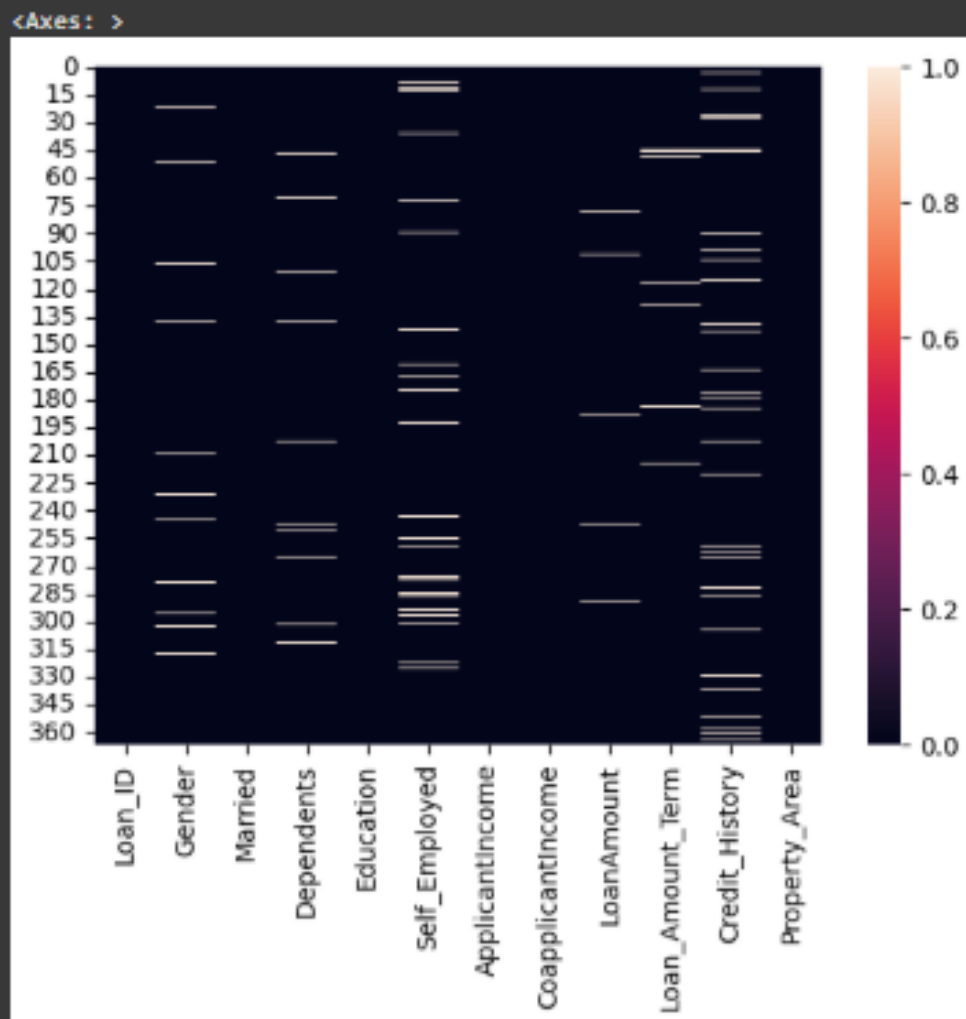
```
df["Loan_Amount_Term"].fillna(df["Loan_Amount_Term"].mean(),inplace=True)
```

```
df["Credit_History"].fillna(1.0,inplace=True)
```

Data Cleaning & Pre-Processing

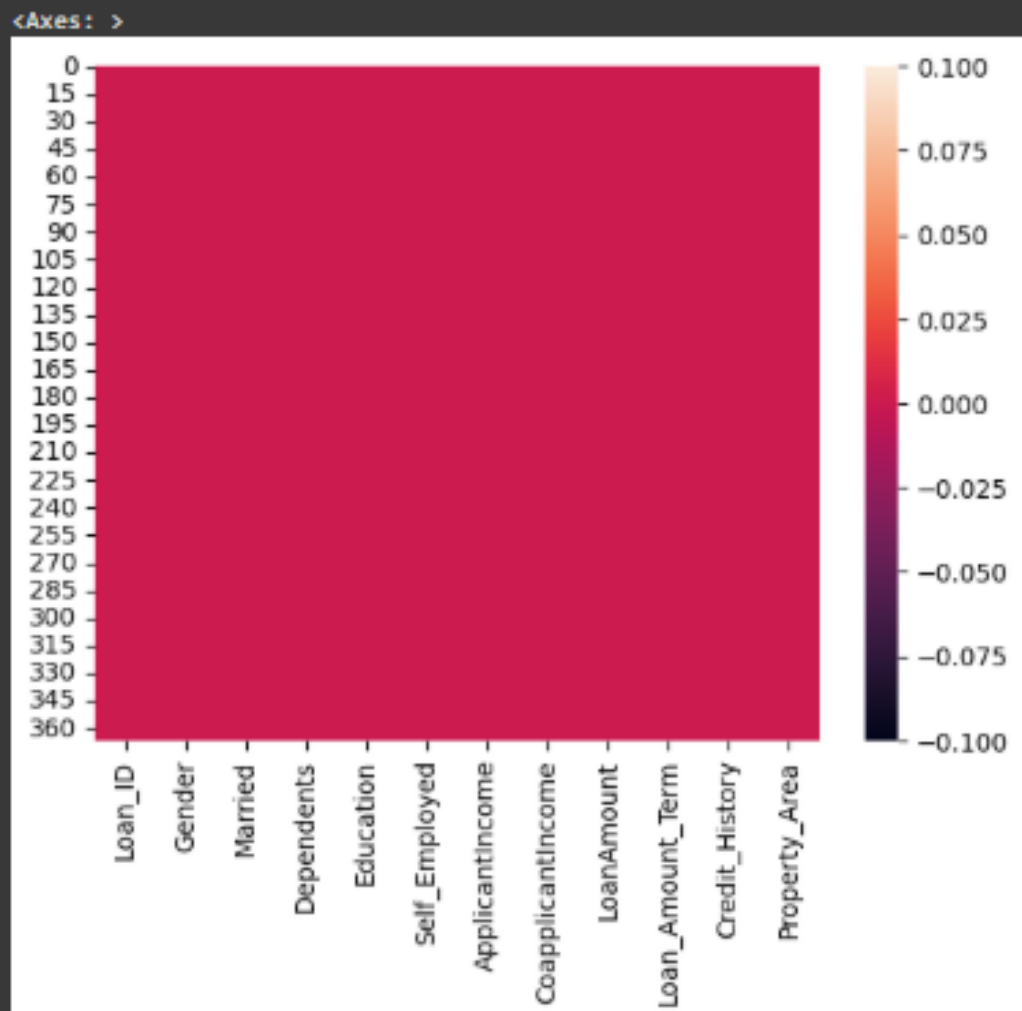
BEFORE CLEANING

```
sns.heatmap(df.isnull())
```



AFTER CLEANING

```
sns.heatmap(df.isnull())
```



- **SUMMARY :** Handling null values and outliers requires a systematic approach that suits the specific characteristics and attributes of the data. By implementing the described strategies, we can effectively manage and fill in missing values, ensuring the dataset's completeness, integrity, and reliability for subsequent analysis and insights.
- With these null, missing, and invalid values properly addressed, we are now prepared to proceed with analyzing the dataset.

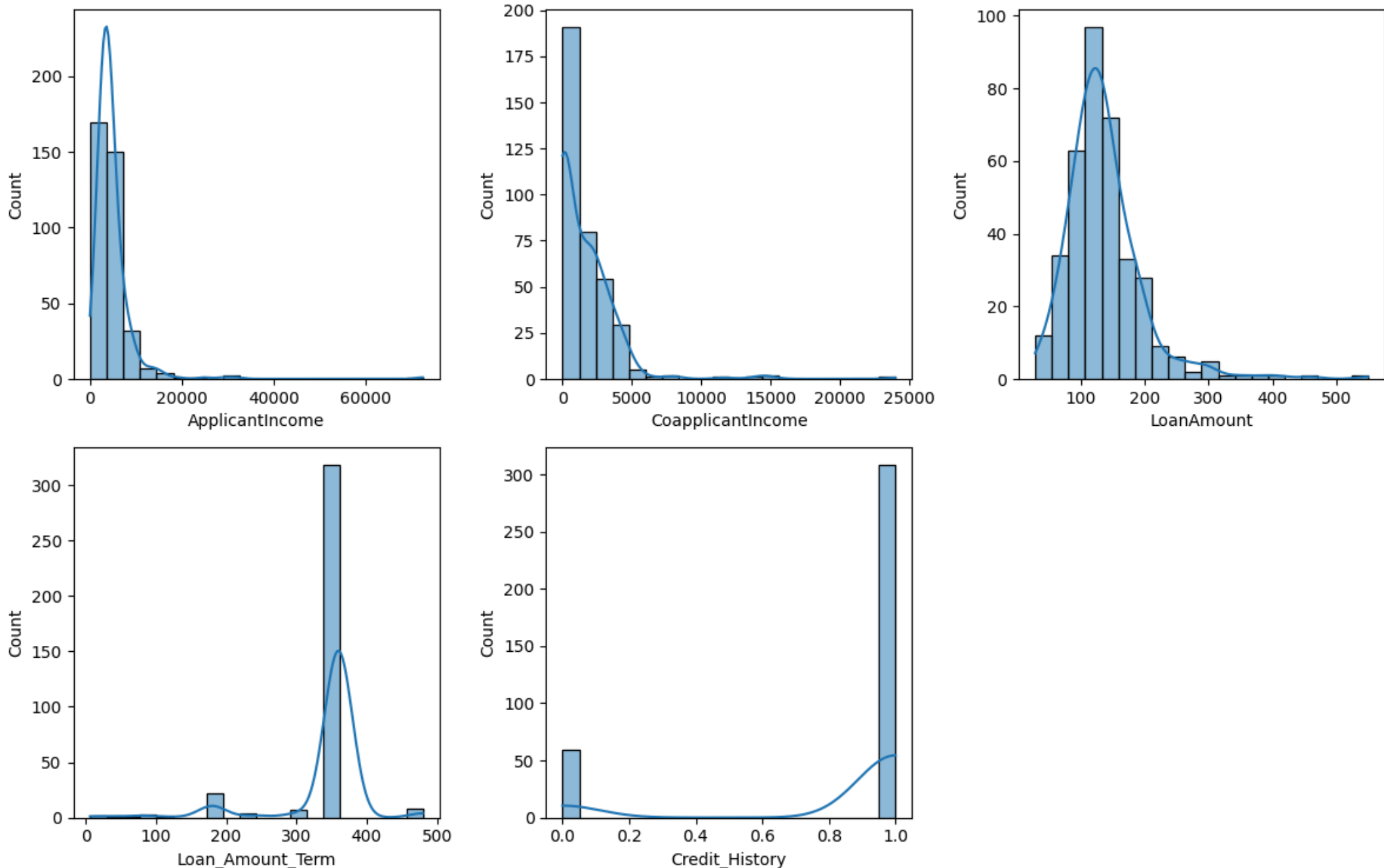
Data Visualization and Insights

Univariate

- **Histograms:** Plot the frequency distribution of key numeric variables.

```
df_1=df.select_dtypes(include=["int64","float64"]).columns.to_list()
plt.figure(figsize=(12,15),facecolor="white")
plotnumber = 1
for i in df_1:
    plt.subplot(4,3,plotnumber)
    sns.histplot(df[i],kde=True,bins=20)
    plt.xlabel(i,fontsize=10)
    plt.ylabel("Count")
    plotnumber += 1
plt.suptitle("HISTOGRAM DISTRIBUTION OF NUMERIC FEATURES")
plt.tight_layout()
plt.show()
```

HISTOGRAM DISTRIBUTION OF NUMERIC FEATURES



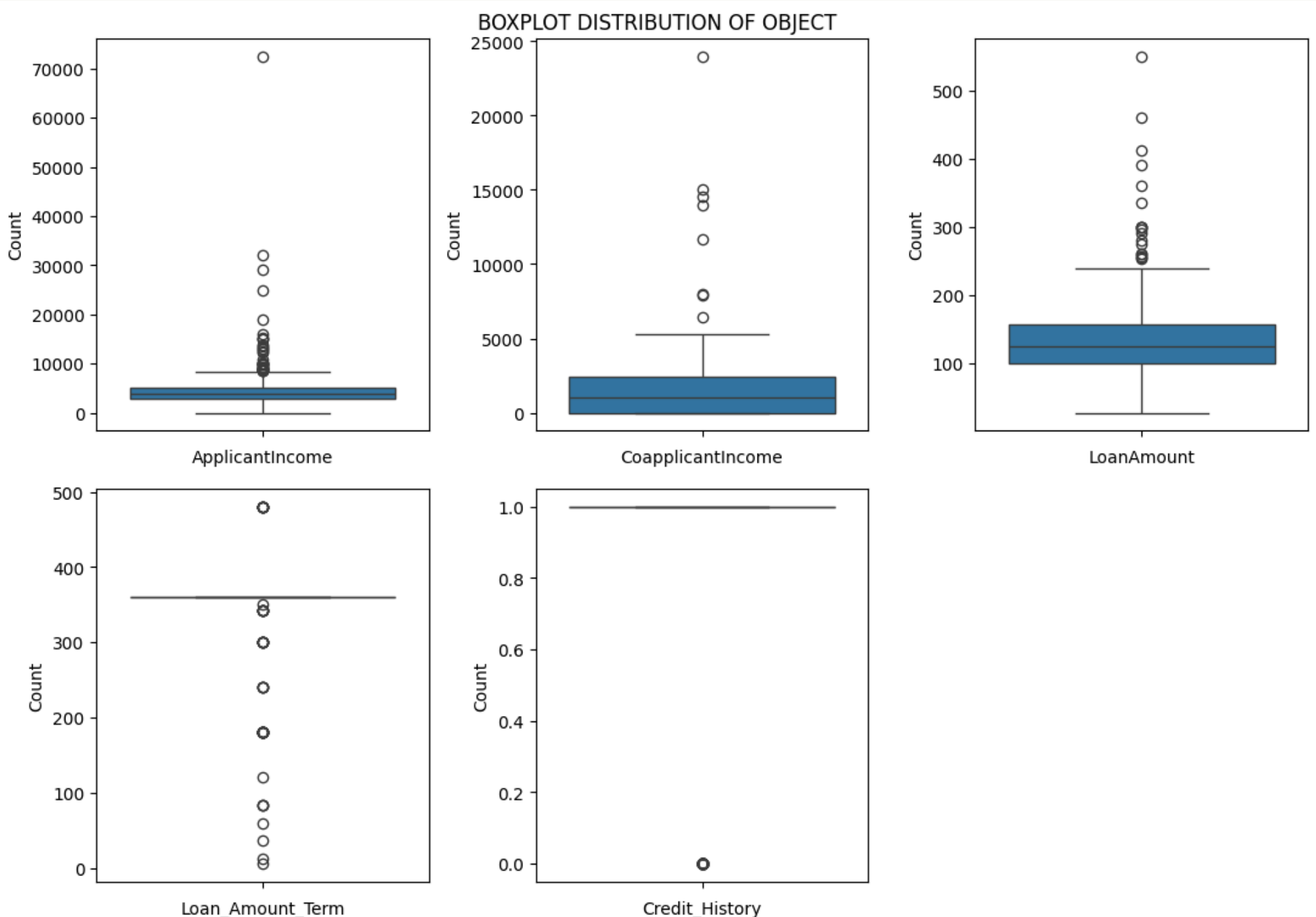
Insights

- **Applicant Income:** The distribution is heavily skewed to the right, with most applicants having a lower income. There is a long tail extending to very high incomes, indicating a few outliers with significantly higher earnings.
- **Coapplicant Income:** This distribution is also right-skewed, but a large number of the entries are at or close to zero, suggesting that many applicants do not have a coapplicant with an income.
- **Loan Amount:** The distribution appears to be somewhat skewed to the right, though less so than the income features. Most loan amounts are concentrated in the lower to middle range, with fewer and fewer loans for higher amounts.
- **Loan Amount Term:** The histogram shows a very strong peak at a specific term, likely a standard loan term like 360 months (30 years). There are also smaller peaks at other common terms. The distribution is not a smooth curve but rather a collection of discrete, high-frequency values.
- **Credit History:** This histogram is bimodal, with a very high bar at a specific value (likely representing "yes" or "1") and a much smaller bar at another value (likely "no" or "0"). This suggests that most applicants have a credit history that meets a certain criterion, but a significant portion do not. The feature is likely a binary or categorical variable, not a truly continuous numeric one, which is why its histogram looks different.

Data Visualization and Insights

- **Box Plots:** Identify potential outliers and visualize the spread of data.

```
df_1=df.select_dtypes(include=["int64","float64"]).columns.to_list()
plt.figure(figsize=(12,15),facecolor="white")
plotnumber = 1
for i in df_1:
    plt.subplot(4,3,plotnumber)
    sns.histplot(df[i],kde=True,bins=20)
    plt.xlabel(i,fontsize=10)
    plt.ylabel("Count")
    plotnumber += 1
plt.suptitle("HISTOGRAM DISTRIBUTION OF NUMERIC FEATURES")
plt.tight_layout()
plt.show()
```



Insights

Applicant Income

The plot for "ApplicantIncome" shows a wide distribution with a significant number of outliers, indicating a large disparity in the income levels of the applicants.

Most of the data is clustered toward the lower end, suggesting that the majority of applicants have a lower income. The median is located near the beginning of the box, which indicates a right-skewed distribution.

The numerous points beyond the upper whisker represent a small number of applicants with very high incomes, which are considered outliers.

Coapplicant Income

The "CoapplicantIncome" plot has a median of 0, which suggests that more than 50% of the loan applicants do not have a co-applicant or that their co-applicant has no income.

The plot also shows a wide range of values and several outliers, indicating that while many applicants have no co-applicant income, a few have a very high co-applicant income.

Loan Amount

The "LoanAmount" plot shows a distribution that is relatively more compact compared to the income plots, but it also has a number of outliers on the higher end. This suggests that while most loans fall within a similar range, there are some unusually large loan amounts.

The median is positioned roughly in the center of the box, indicating that the loan amounts for the middle 50% of applicants are more symmetrically distributed than the income values.

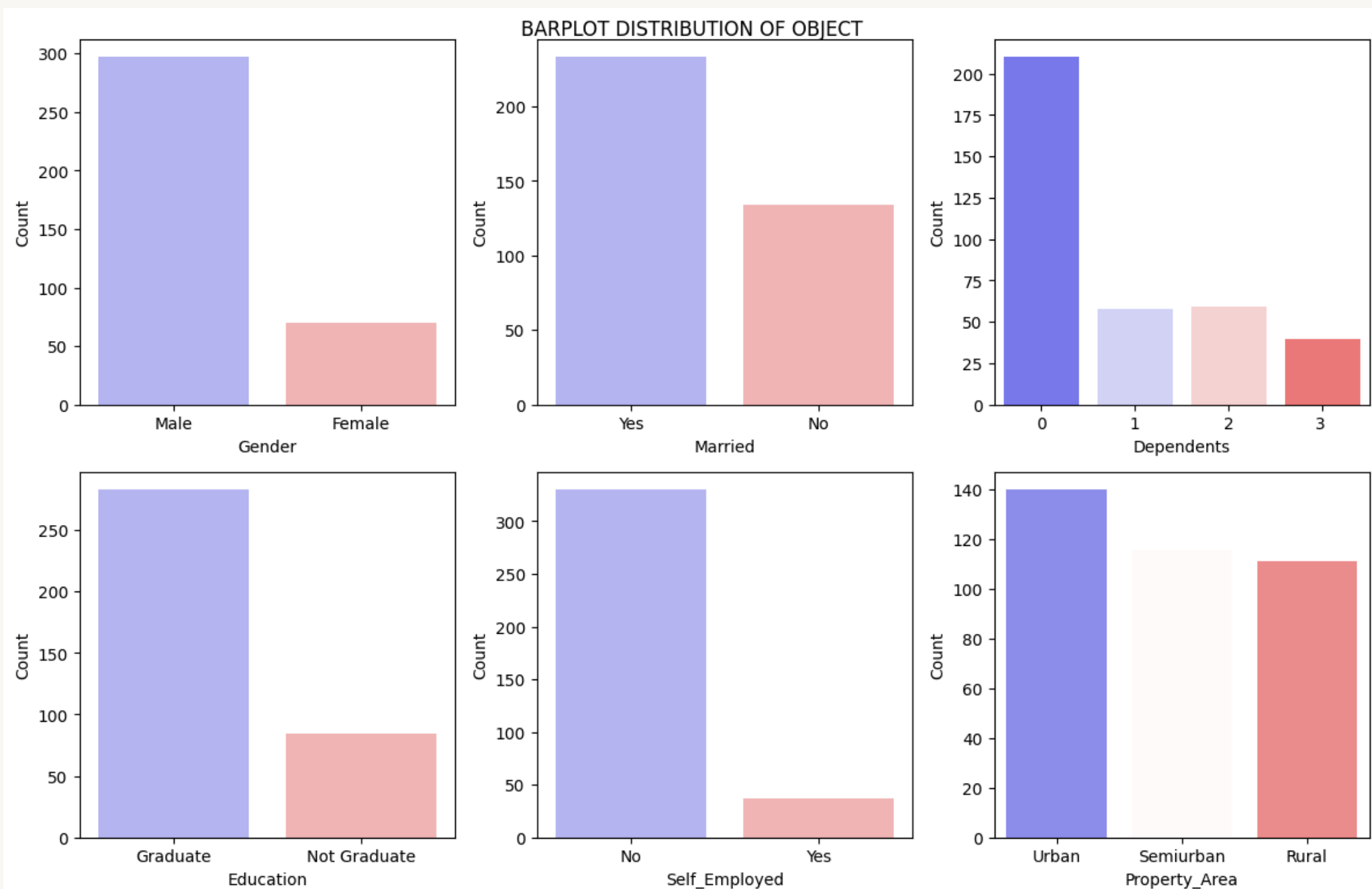
Credit History

The "Credit_History" plot is a categorical variable represented by a box plot, which is less common but can still show distribution. The plot suggests that most applicants have a credit history score concentrated around a single value, likely representing a good credit history, as this is a key factor in loan approval.

Data Visualization and Insights

- **Bar Charts:** Visualize the frequency distribution of categorical variables.

```
df_2=df.select_dtypes(include="object")
df_2.drop(columns=["Loan_ID"],inplace=True)
plt.figure(figsize=(19,15),facecolor="white")
plotnumber = 1
for i in df_2:
    plt.subplot(4,3,plotnumber)
    sns.countplot(x=df[i],palette="bwr")
    plt.xlabel(i,fontsize=10)
    plt.ylabel("Count")
    plotnumber += 1
plt.suptitle("BARPLOT DISTRIBUTION OF OBJECT")
plt.tight_layout()
plt.show()
```



Insights

Gender: There are a significantly higher number of male applicants than female applicants.

Married: The majority of the applicants are married.

Dependents: Most applicants have zero dependents, and a small number have three or more.

Education: The number of applicants with a graduate-level education is much higher than those who are not graduates.

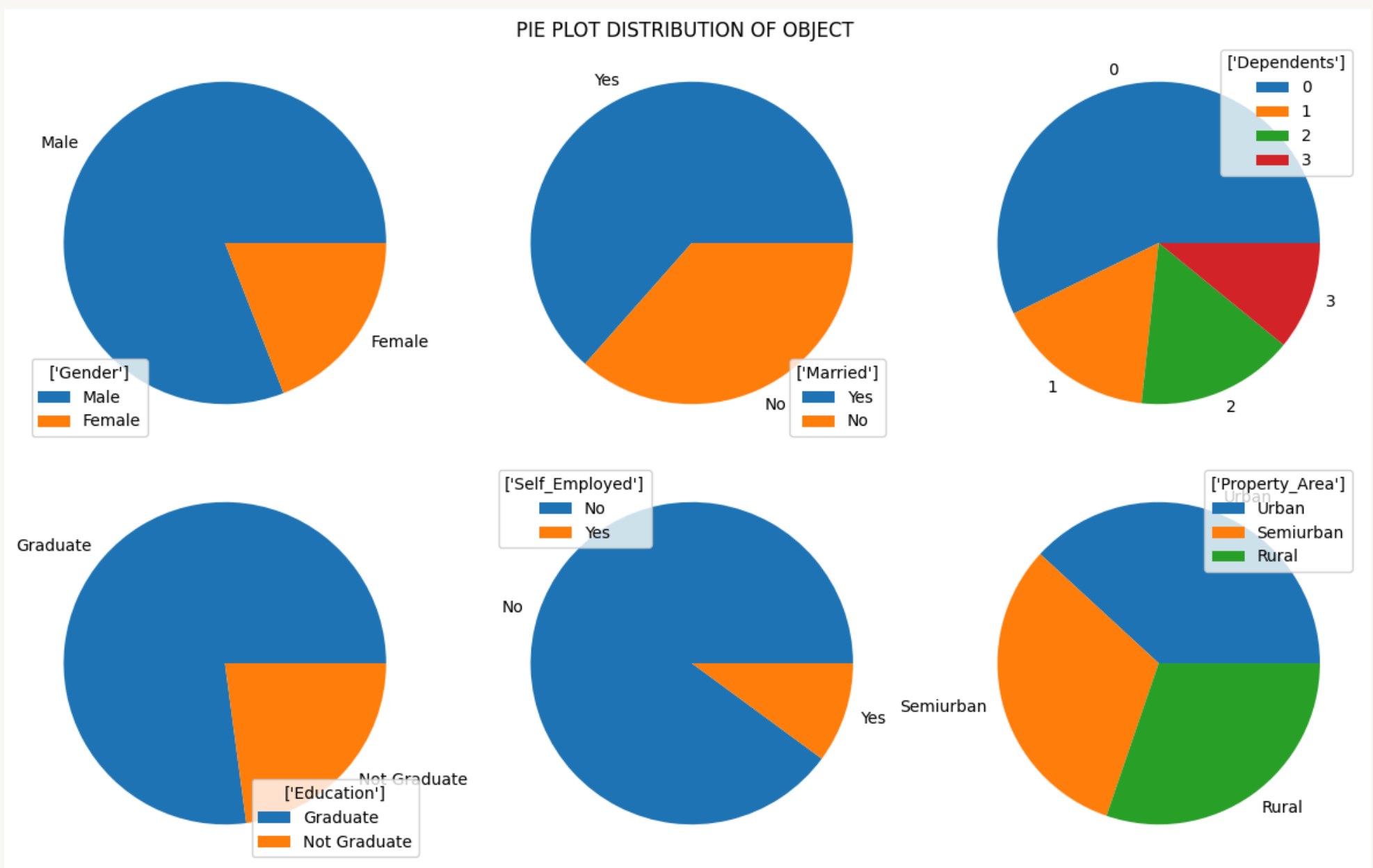
Self Employed: The majority of the applicants are not self-employed.

Property Area: The highest number of applicants are from semiurban and urban areas, while the lowest number are from rural areas.

Data Visualization and Insights

- **Pie Charts:** Represent the composition of categorical variables.

```
df_2=df.select_dtypes(include="object")
df_2.drop(columns=["Loan_ID"],inplace=True)
plt.figure(figsize=(12,15),facecolor="white")
plotnumber = 1
for i in df_2:
    plt.subplot(4,3,plotnumber)
    plt.pie(df[i].value_counts(),labels=df[i].unique())
    plt.legend(title=[i])
    plotnumber +=1
plt.suptitle("PIE PLOT DISTRIBUTION OF OBJECT")
plt.tight_layout()
plt.show()
```



Insights

Gender: The pie chart for "Gender" shows a clear majority of the dataset is male, with the blue slice being significantly larger than the orange slice for female.

Education: The "Education" chart indicates that a large majority of individuals are "Graduate" (blue slice), while a smaller portion is "Not Graduate" (orange slice).

Property Area: The distribution across property areas appears to be relatively even, with the "Semiurban" and "Urban" slices being a similar size and the "Rural" slice being slightly larger.

Dependents: The "Dependents" chart shows a clear hierarchy, with the largest group having "0" dependents (blue slice), followed by "1" dependent (orange slice), and then smaller proportions for "2" and "3" or more dependents.

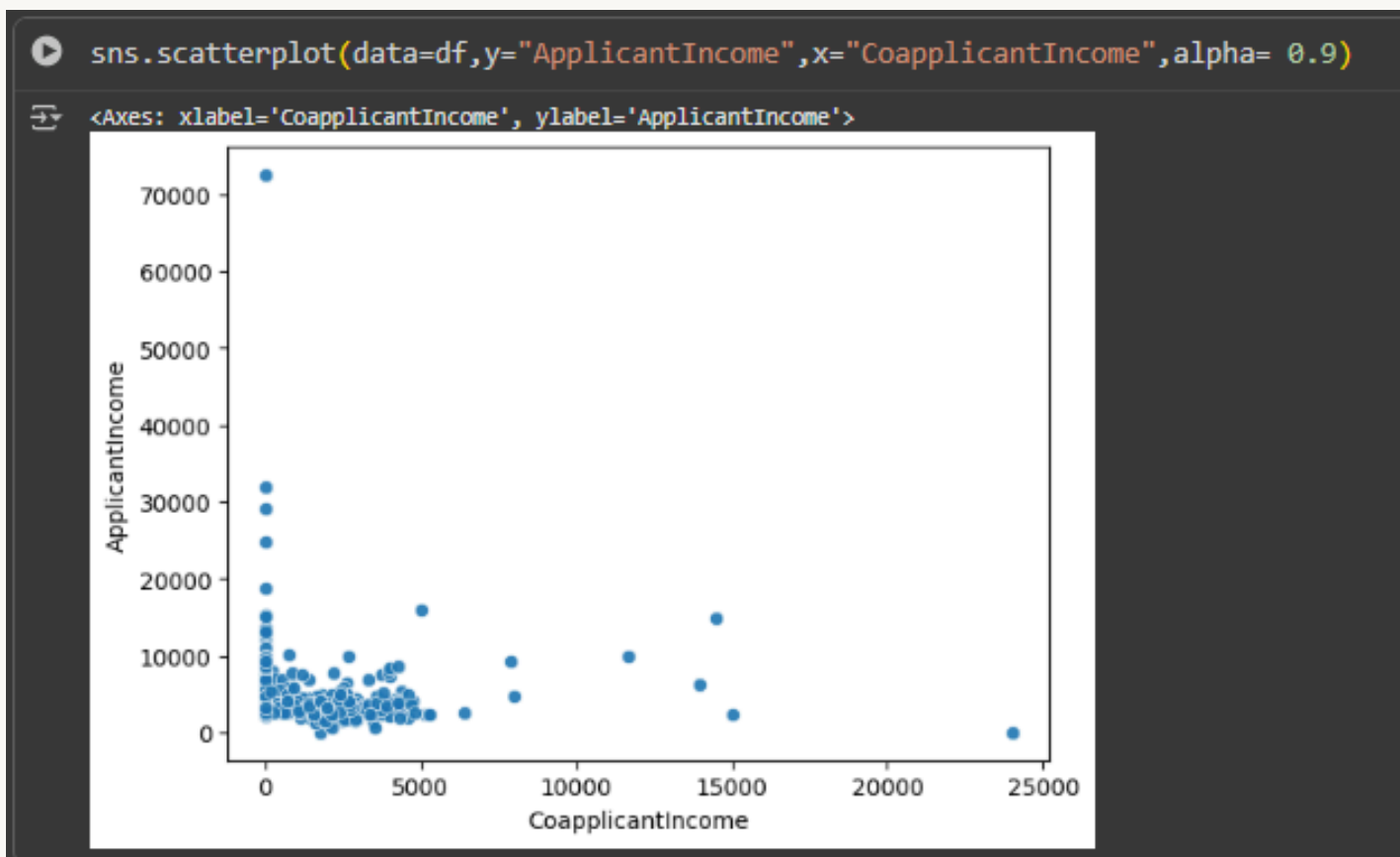
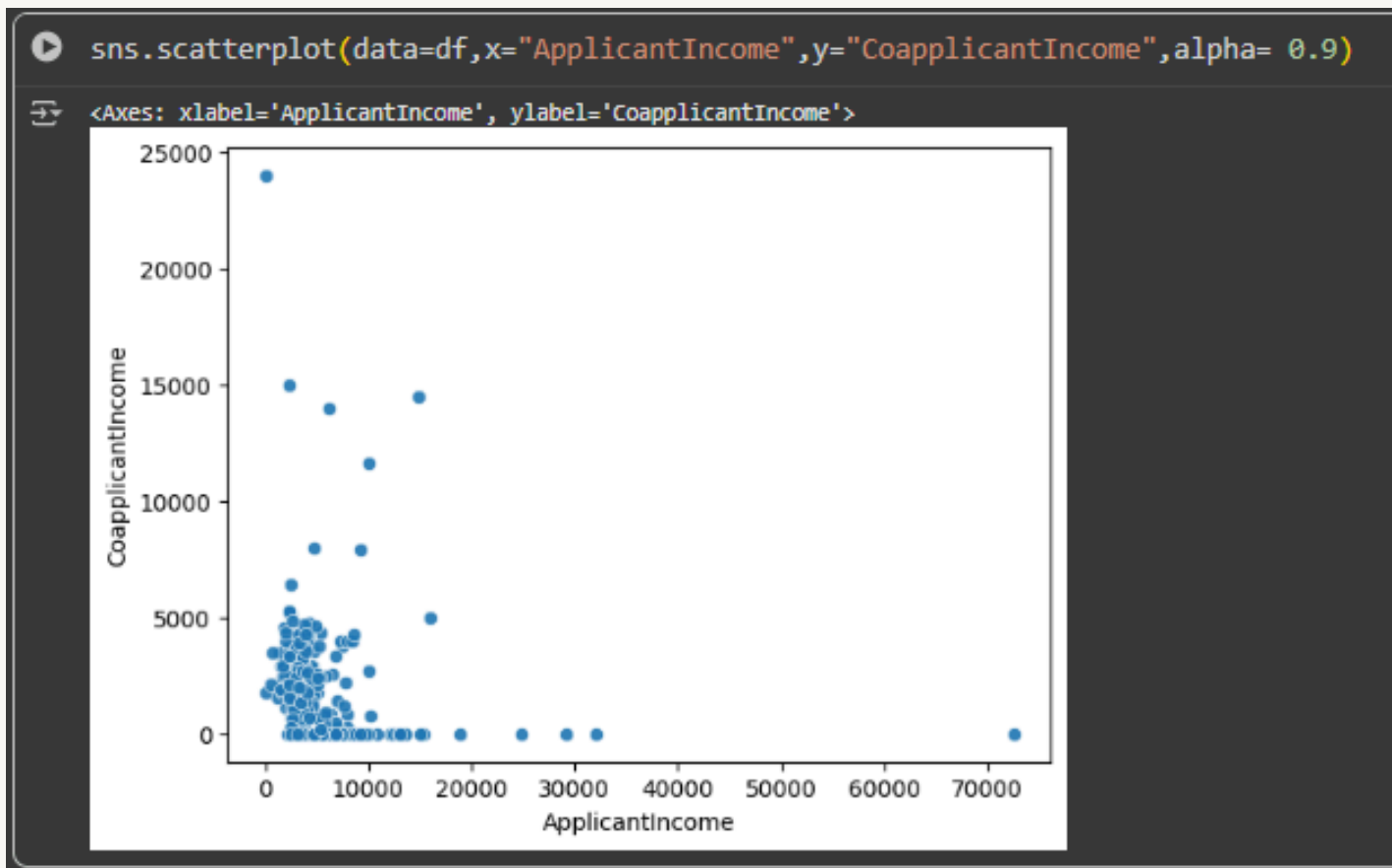
Marital Status: The chart for "Married" suggests that a larger portion of the dataset is married (blue slice) compared to those who are not (orange slice).

Self-Employed: The "Self_Employed" chart indicates that the majority of the population is not self-employed (blue slice), while a smaller portion is (orange slice).

Data Visualization and Insights

Bivariate

- Create scatter plots to explore relationships between pairs of numeric variables.



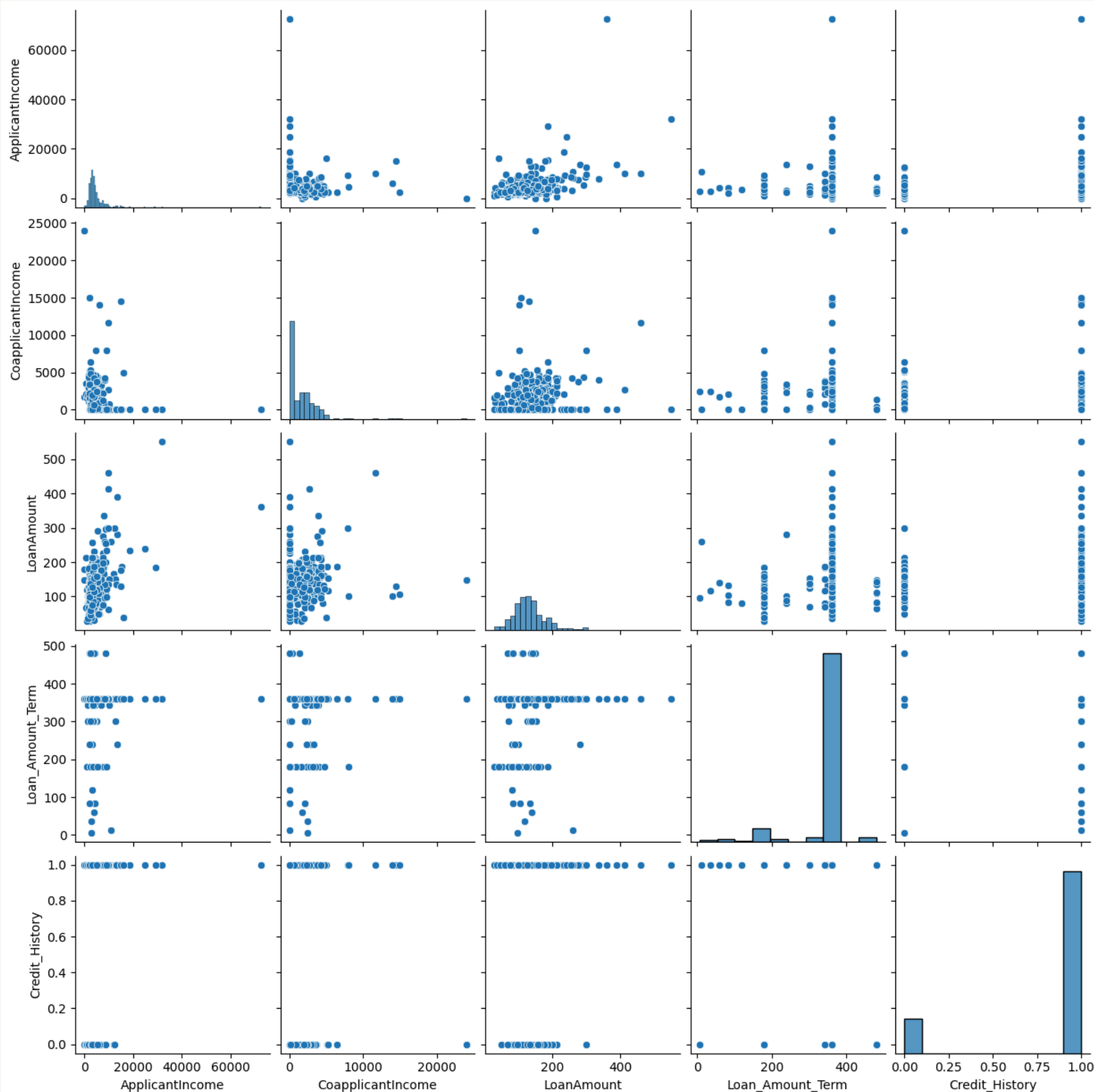
Insights

Dominant Group: A significant number of data points are clustered along the y-axis where Coapplicantincome is zero. This suggests that for a large portion of the individuals in this dataset, either there is no co-applicant or the co-applicant has no income.

Lack of Strong Correlation: The plot does not show a strong positive or negative correlation between the two variables. This indicates that a person's income level (ApplicantIncome) is not directly predictive of their co-applicant's income level, and vice-versa.

Data Visualization and Insights

- Use pair plots (scatter matrix) to visualize interactions between multiple numeric variables simultaneously.

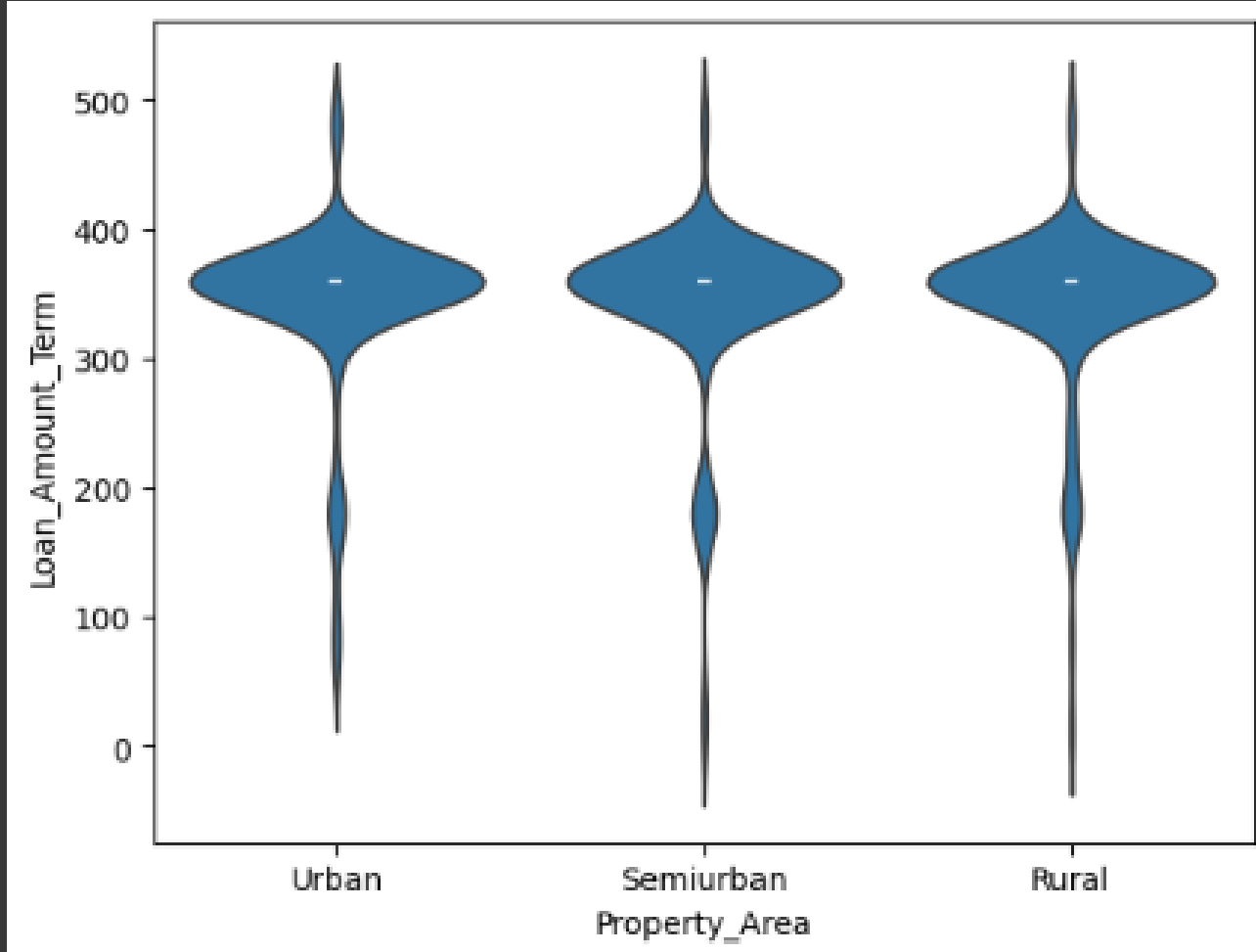


Data Visualization and Insights

- Investigate the relationship between categorical and numeric variables using box plots or violin plots.

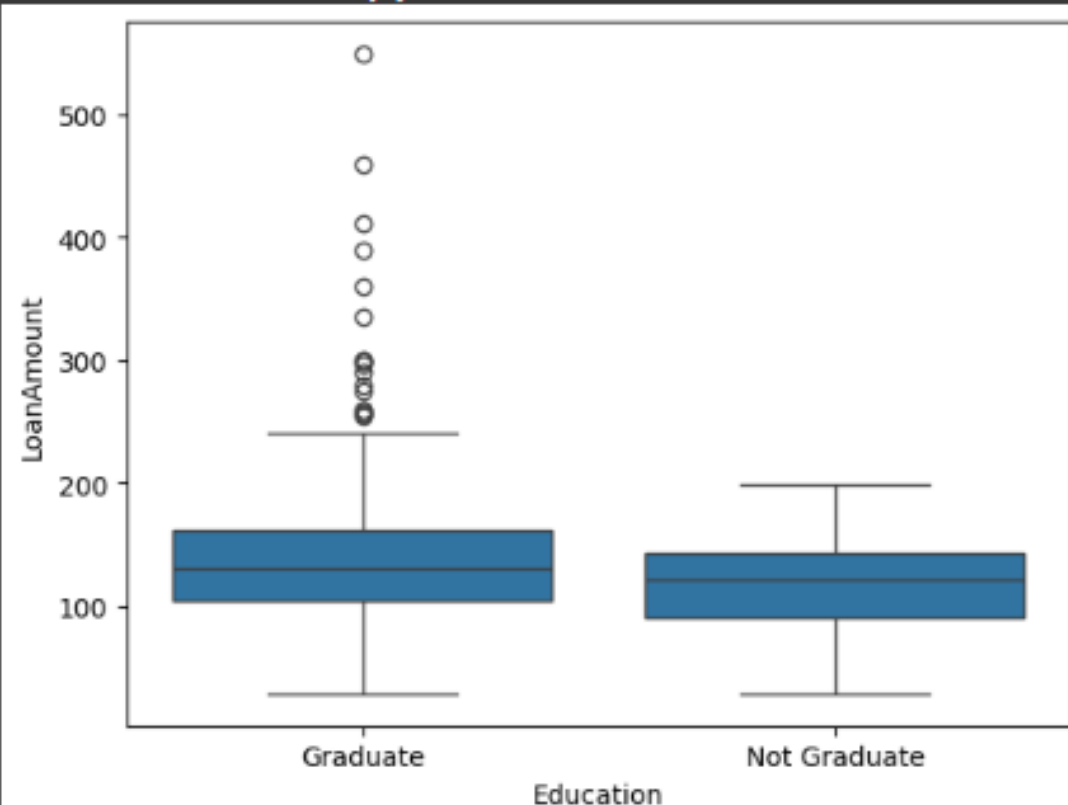
```
sns.violinplot(data=df,x="Property_Area",y="Loan_Amount_Term")
```

```
<Axes: xlabel='Property_Area', ylabel='Loan_Amount_Term'>
```



```
sns.boxplot(data=df,x="Education",y="LoanAmount")
```

```
<Axes: xlabel='Education', ylabel='LoanAmount'>
```



Insights

Insights from the Violin Plot

The violin plot visualizes the distribution of Loan_Amount_Term across different Property_Area categories. The width of the violin shape at any given point represents the density of data points.

The most significant insight is that the majority of loans for all three property areas—Urban, Semiurban, and Rural—have a term of around 360 months. This is indicated by the very wide section of each violin around that value.

While the 360-month term is dominant across all property areas, the distributions for Semiurban and Rural areas appear slightly more spread out, suggesting a bit more variability in loan terms for these areas compared to Urban.

Insights from the Box Plot

The box plot compares the distribution of LoanAmount between two groups: "Graduate" and "Not Graduate".

The median LoanAmount (the line inside the box) is nearly identical for both groups, falling somewhere between 100 and 200 on the y-axis.

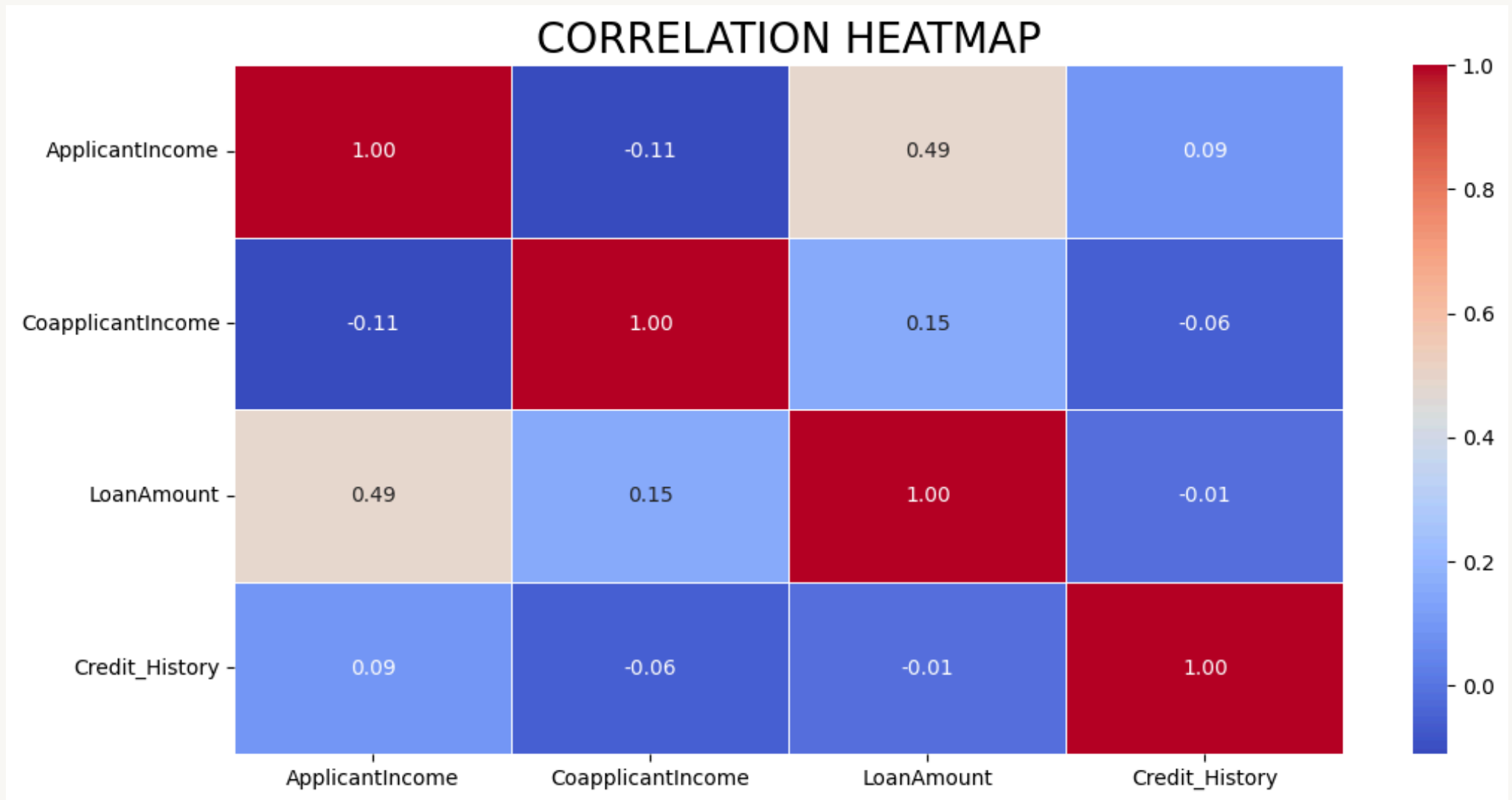
The spread of the middle 50% of the data, represented by the box itself, is also similar for both groups, suggesting that the variability in loan amounts is comparable regardless of education level.

The dots above the boxes represent outliers. This shows that a few individuals in both the "Graduate" and "Not Graduate" categories have received loan amounts that are much higher than the rest of their respective groups.

Data Visualization and Insights

- Visualize correlations using a heatmap.

Multivariate



Insights from Correlation Heatmap

ApplicantIncome and LoanAmount have a positive correlation, indicating that higher applicant incomes tend to be associated with larger loan amounts. This aligns with the intuitive understanding that borrowers with higher incomes request larger loans, possibly due to higher repayment capacity.

CoapplicantIncome shows a moderate positive correlation with LoanAmount, suggesting that when there is a coapplicant with higher income, the total loan amount requested tends to be bigger. This highlights the combined earning potential's role in loan size determination.

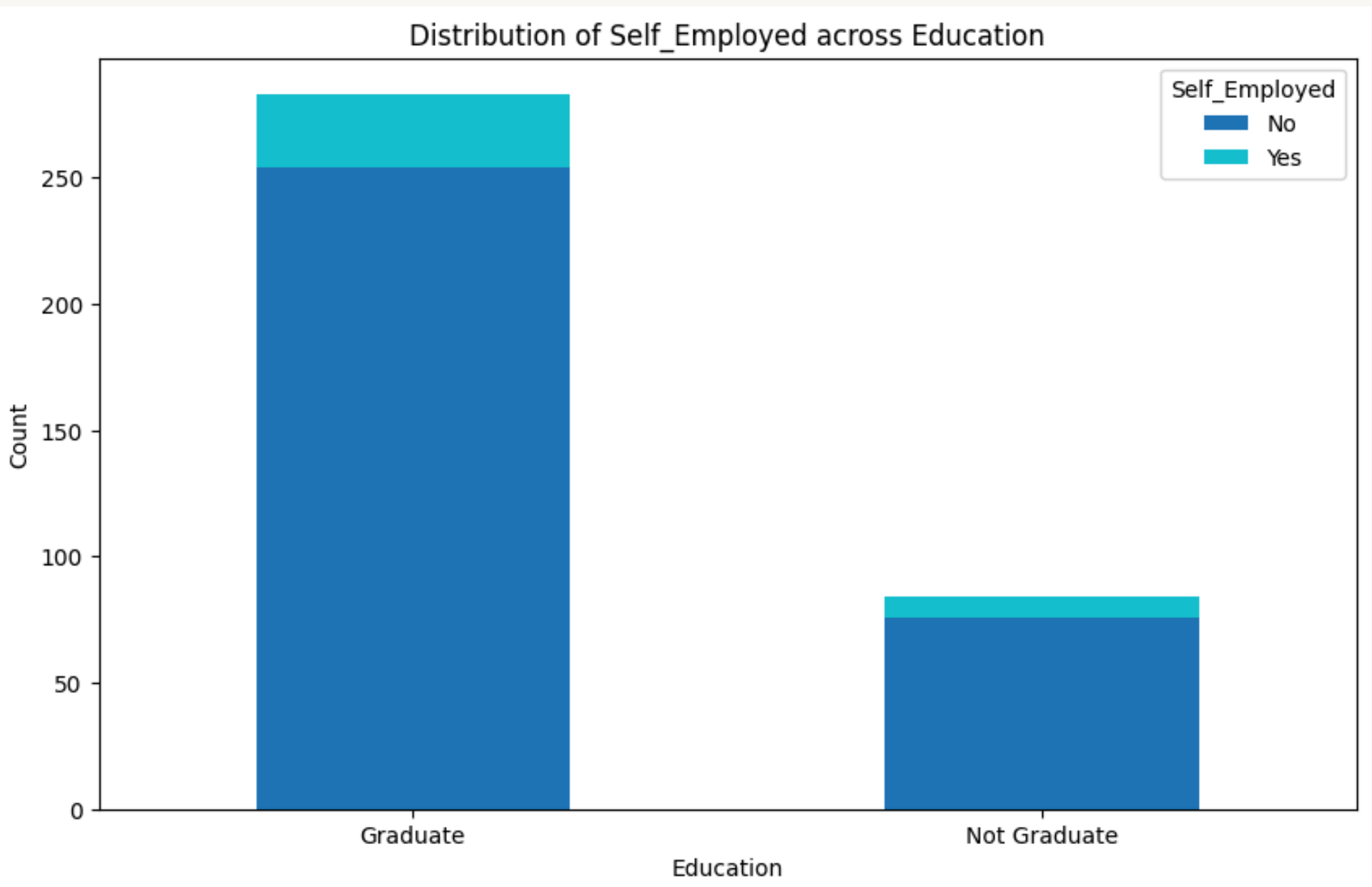
The correlation between ApplicantIncome and CoapplicantIncome is very low or near zero, which suggests that the incomes of primary applicants and coapplicants vary independently in this dataset.

Credit_History has weak or negligible correlation with the other numeric variables, reflecting that this binary or categorical feature (often 0 or 1) may not directly move in tandem with continuous income or loan size variables. However, it remains vital for approval decisions as a risk indicator.

Data Visualization and Insights

- Create a stacked bar chart to show the distribution of categorical variables across multiple categories

```
pd.crosstab(df['Education'], df['Self_Employed']).plot(  
    kind='bar', stacked=True, figsize=(10,6), colormap='tab10'  
)  
plt.title('Distribution of Self_Employed across Education')  
plt.xlabel('Education')  
plt.ylabel('Count')  
plt.xticks(rotation=0)  
plt.legend(title='Self_Employed')  
plt.show()
```



Insights

The stacked bar chart depicting the distribution of Self_Employed applicants across different education levels indicates that individuals with higher education are more likely to be self-employed. The chart shows a visibly larger segment of self-employed individuals within the educated group, suggesting a strong association between education and self-employment status. Conversely, less educated applicants tend to be predominantly salaried or not self-employed, highlighting a possible link between higher education and entrepreneurial or self-employed pursuits. This insight could be useful for understanding employment patterns among borrowers and tailoring financial products or risk assessments based on education and employment status.

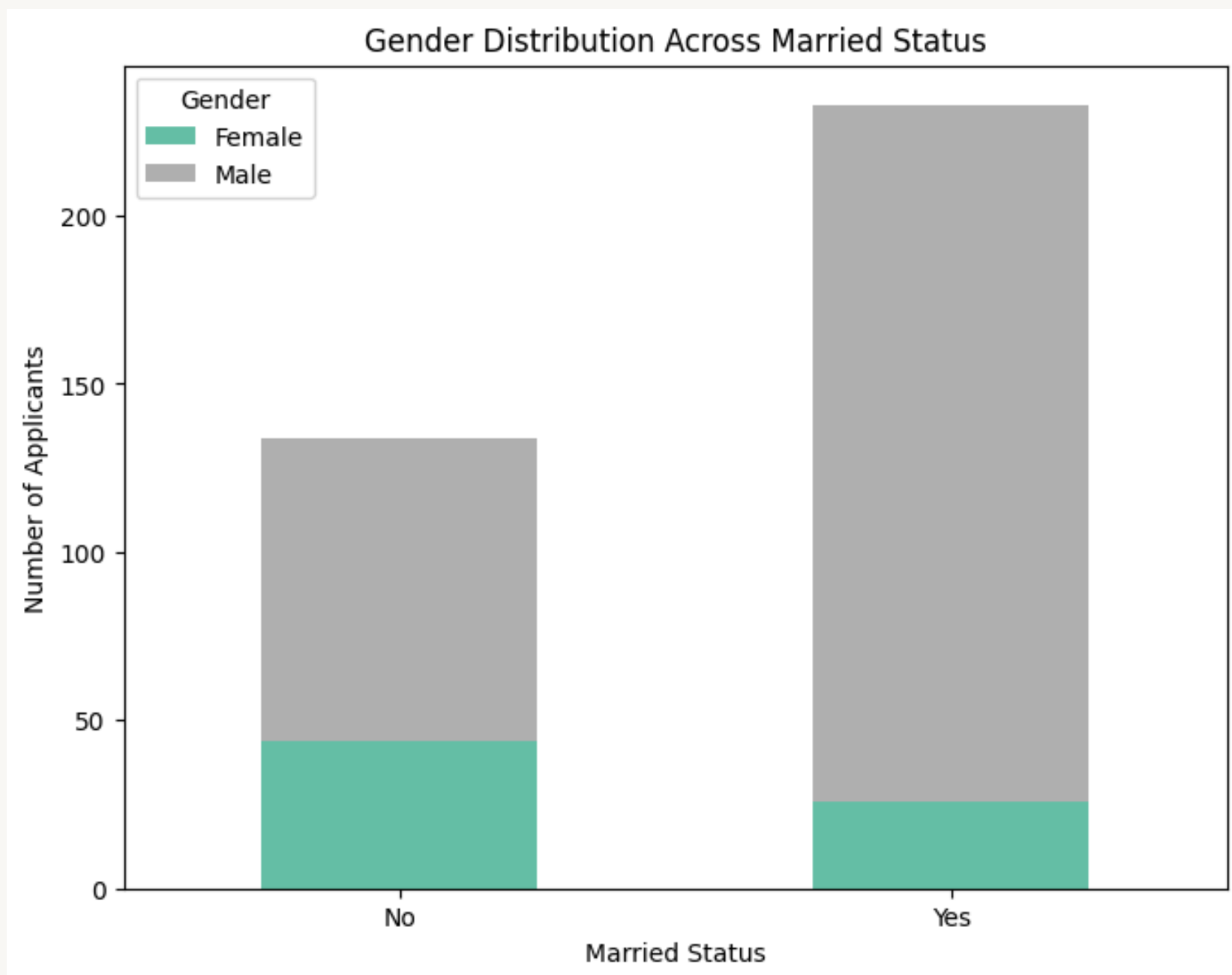
Data Visualization and Insights

- Create a stacked bar chart to show the distribution of categorical variables across multiple categories

```
ct = pd.crosstab(df['Married'], df['Gender'])

ct.plot(kind='bar', stacked=True, figsize=(8,6), colormap='Set2')

plt.title('Gender Distribution Across Married Status')
plt.xlabel('Married Status')
plt.ylabel('Number of Applicants')
plt.xticks(rotation=0)
plt.legend(title='Gender')
plt.show()
```



Insights

The stacked bar chart illustrating Gender distribution across Married status reveals distinct demographic patterns in the loan applicant pool. It shows that a significantly larger proportion of married applicants are male, indicating that males are more likely to be married loan applicants. Conversely, among unmarried applicants, while males still dominate, the proportion of female applicants is comparatively higher than in the married group.

FINAL REPORT

❑CONCLUSIONS:

- Income and Loan Amount: Applicant income significantly influences the amount of the loan they can receive.
 - Demographic Factors: Factors such as gender, marital status, education, and employment status impact the characteristics of loan applications.
 - Credit History: The credit history of applicants affects their loan application outcomes
- FINAL REPORT Summarizing the key findings, drawing conclusions, and offering recommendations based on the insights obtained from the analysis:

❑Recommendations :

- Target Marketing: Customize loan products and marketing strategies according to income levels and demographic characteristics.
- Risk Assessment: Evaluate income, credit history, and other factors to inform loan approval and risk assessment.
- Product Diversification: Provide loan products with different terms and amounts to meet diverse customer needs.
- Further Analysis: Investigate additional factors and their interactions to enhance insights and improve decision-making.

Thanks for reading

for coding part.....

https://colab.research.google.com/drive/1ucZg7ezkroh5IMbjszlvzsUNtiSw65ho#scrollTo=Sq5syu3e_E_l

